UNF

PartsOrder
customerNo VARCHAR
customerName VARCHAR
customerType VARCHAR
orderDate DATE
orderTime TIME
employeeName VARCHAR
partNo INTEGER
partName VARCHAR
partType VARCHAR
cageCode VARCHAR
quantityOrdered INTEGER
unitPrice DECIMAL

1NF

PartsOrder	
PK	**customerNo VARCHAR**
PK	**orderDate DATE**
PK	**orderTime TIME**
PK	**partNo INTEGER**
	customerName VARCHAR
	customerType VARCHAR
	employeeName VARCHAR
	partName VARCHAR
	partType VARCHAR
	cageCode VARCHAR
	quantityOrdered INTEGER
	unitPrice DECIMAL

2NF

Customer	
PK	**customerNo VARCHAR**
	customerName VARCHAR
	customerType VARCHAR

Part	
PK	**partNo INTEGER**
	partName VARCHAR
	partType VARCHAR
	cageCode VARCHAR
	unitPrice DECIMAL

Order	
PK	**customerNo VARCHAR**
PK	**orderDate DATE**
PK	**orderTime TIME**
	employeeName VARCHAR

OrderPart	
PK	**customerNo VARCHAR**
PK	**orderDate DATE**
PK	**orderTime TIME**
PK	**partNo INTEGER**
	quantityOrdered INTEGER

3NF

Customer	
PK	**customerNo VARCHAR**
	customerName VARCHAR
	customerType VARCHAR

Part	
PK	**partNo INTEGER**
	partName VARCHAR
	partType VARCHAR
	cageCode VARCHAR
	unitPrice DECIMAL

Order	
PK+FK	***customerNo VARCHAR***
PK	**orderDate DATE**
PK	**orderTime TIME**
	employeeName VARCHAR

OrderPart	
PK+FK	***customerNo VARCHAR***
PK+FK	***orderDate DATE***
PK+FK	***orderTime TIME***
PK+FK	***partNo INTEGER***
	quantityOrdered INTEGER

Question 1

A. Assumptions

- Each customer has a unique customer number.
- Each part has a unique part number.
- Customer number, order date, and order time together uniquely identify an order.
- Each part appears only once per order.
- Each part has one name, type, cage code, and unit price.
- Employees can assist any customer.

B. First Normal Form (1NF)

The original form contains repeating groups because multiple parts can be included in one order. To achieve 1NF, each ordered part is placed in a separate row, and every attribute contains a single value.

Primary Key: (customerNo, orderDate, orderTime, partNo)

C. Second Normal Form (2NF)

Partial dependencies are removed by separating customer information, part information, and order information into different tables.

Relations: Customer, Part, Order, OrderPart.

D. Third Normal Form (3NF)

The relations already satisfy 3NF because no transitive dependencies remain under the stated assumptions. Therefore, no additional tables are required.

Final Relations: Customer, Part, Order, OrderPart.

The foreign keys connect customers to orders and orders to their individual parts.

UNF

AppointmentRecord

	staffNo VARCHAR
	therapistName VARCHAR
	patNo VARCHAR
	patName VARCHAR
	appointmentDate DATE
	appointmentTime TIME
	branchNo VARCHAR

1NF

AppointmentRecord

PK	**staffNo VARCHAR**
PK	**appointmentDate DATE**
PK	**appointmentTime TIME**
	therapistName VARCHAR
	patNo VARCHAR
	patName VARCHAR
	branchNo VARCHAR

2NF

Therapist

PK	**staffNo VARCHAR**
	therapistName VARCHAR

TherapistSchedule

PK	**staffNo VARCHAR**
PK	**appointmentDate DATE**
	branchNo VARCHAR

Appointment

PK	**staffNo VARCHAR**
PK	**appointmentDate DATE**
PK	**appointmentTime TIME**
	patNo VARCHAR
	patName VARCHAR

3NF

Therapist

PK	**staffNo VARCHAR**
	therapistName VARCHAR

Patient

PK	**patNo VARCHAR**
	patName VARCHAR

TherapistSchedule

PK,FK	***staffNo VARCHAR***
PK	**appointmentDate DA**TE
	branchNo VARCHAR

Appointment

PK,FK	***staffNo VARCHAR***
PK,FK	***appointmentDate DATE***
PK	**appointmentTime TIME**
FK	*patNo VARCHAR*

Question 2

A. Assumptions

- Each therapist has a unique staff number.
- Each patient has a unique patient number.
- A therapist can work at multiple branches but only one branch per day.
- Patients can have multiple appointments with different therapists.
- Each therapist can have only one appointment at a specific date and time.
- Each appointment involves one therapist and one patient.

B. First Normal Form (1NF)

The original table already satisfies 1NF because every attribute contains a single value and there are no repeating groups.

Primary Key: (staffNo, appointmentDate, appointmentTime)

C. Second Normal Form (2NF)

Partial dependencies are removed by separating therapist details and daily branch assignments from appointment information.

Relations: Therapist, TherapistSchedule, Appointment.

D. Third Normal Form (3NF)

The transitive dependency between patNo and patName is removed by creating a separate Patient table.

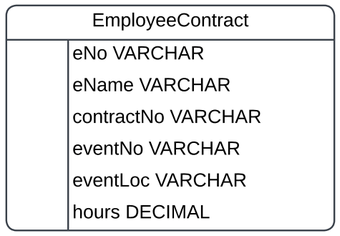
Final Relations: Therapist, Patient, TherapistSchedule, Appointment.

Foreign Keys:

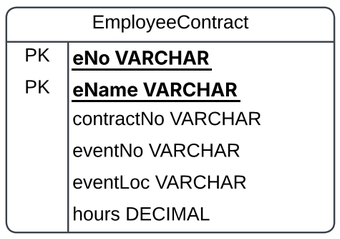
- TherapistSchedule.staffNo references Therapist.staffNo.
- Appointment.(staffNo, appointmentDate) references TherapistSchedule.(staffNo, appointmentDate).
- Appointment.patNo references Patient.patNo.

The final relations satisfy 3NF because all partial dependencies have been removed.

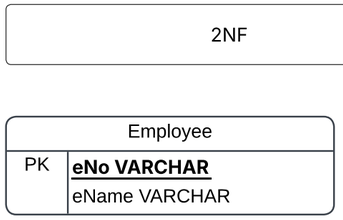
UNF



1NF



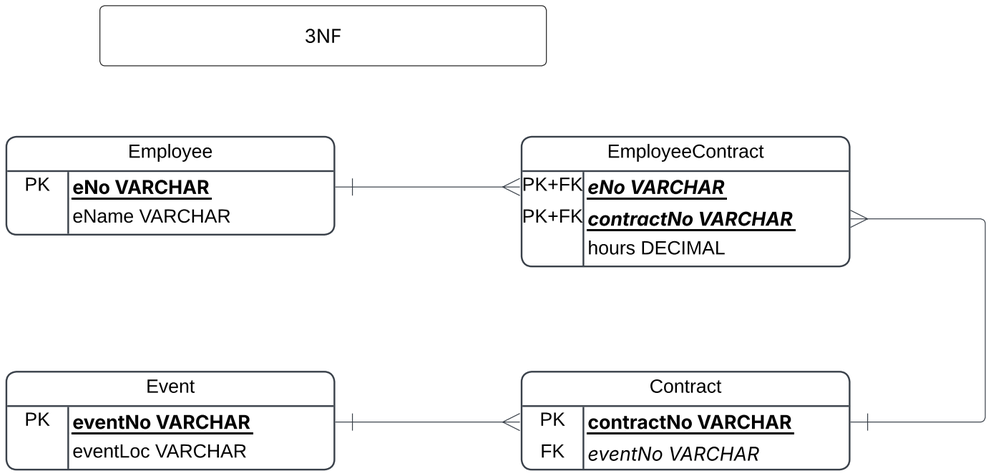
2NF







3NF



Question 3 – Employee Contract Normalization

A. Assumptions

- Each employee has a unique employee number (eNo).
- Each contract has a unique contract number (contractNo).
- Each contract belongs to only one event.
- An event can have multiple contracts.
- Each employee can work on multiple contracts.
- Each event has one recorded location.
- Hours represent the total hours an employee works on a particular contract.

B. First Normal Form (1NF)

The original table already satisfies 1NF because all attributes contain atomic values and there are no repeating groups.

Primary Key: (eNo, contractNo)

C. Second Normal Form (2NF)

Partial dependencies are removed by separating employee information and contract details into different tables.

Relations: Employee, Contract, EmployeeContract.

D. Third Normal Form (3NF)

The transitive dependency between contractNo, eventNo, and eventLoc is removed by creating a separate Event table.

Final Relations: Employee, Event, Contract, EmployeeContract.

Foreign Keys:

- Contract.eventNo references Event.eventNo.
- EmployeeContract.eNo references Employee.eNo.
- EmployeeContract.contractNo references Contract.contractNo.

The final relations satisfy 3NF because all partial dependencies have been eliminated.